

GO-6+ Next Ultimate reality

Methodological treatise / scientific working paper — an extended Physical-AI benchmark for robust decision-making

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REAL-AI-Benchmark project

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Abstract

GO-6+ Next Ultimate reality formalizes a stricter class of Physical-AI benchmarks. The test evaluates whether a multimodal model can read analog instruments, separate relevant inputs from decoy indicators, apply calibration, perform fuzzy risk assessment, simulate nominal and pessimistic system dynamics, select a robust control action, and reproduce the same procedure through an executable Zig verifier. Unlike benchmarks that start from digitally prepared inputs, GO-6+ introduces measurement intervals, raw/corrected separation, a nonlinear vibration channel, physical flow clamping, a control-voltage signal that simultaneously supports pressure relief and increases vibration stress, and a strict formal JSON trace. The methodology is designed to evaluate the full chain perception → measurement → inference → simulation → decision → verification, rather than only the final numeric answer.

Keywords: *Physical-AI benchmarking, LMM, analog instruments, fuzzy logic, robust decision-making, dynamic simulation, Zig verification, measurement intervals, decoy instruments.*

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1. Introduction and research motivation

GO-6+ Next Ultimate reality was created to move multimodal model evaluation beyond static, text-prepared problems and closer to engineering situations in which the input is not an ideal digital JSON object, but a physical reading. In such a setting an error does not arise only in arithmetic. It may arise in reading a scale, recognizing a unit, distinguishing a relevant instrument from a decoy, applying a correction, understanding the system dynamics, or writing a verifier that fails to reproduce the text solution.

The central idea of GO-6+ is to evaluate the complete technical decision chain. The model must see the instrument, understand what it measures, estimate the value and interval, apply calibration, construct the system input vector, perform fuzzy inference, simulate the consequences of multiple actions, and select a robust decision.

GO-6+ is not a safety-certified industrial controller and does not claim to be a complete model of a real plant. Its purpose is methodological: to expose whether a model can preserve discipline across perceptual, numerical, causal and programmatic layers of the same problem.

2. From GO-6 to GO-6+: the methodological change

GO-6 Rev. 1.1 already introduced anti-leak mapping of analog instruments, fuzzy logic, dynamic simulation, and Zig verification. GO-6+ adds additional layers that make the task substantially harder without merely increasing text length.

Layer	GO-6 Rev. 1.1	GO-6+ Next
Number of instruments	5 relevant instruments	9 instruments: 6 relevant + 3 decoys
Mapping	anti-leak, but without a decoy layer	hidden mapping + decoy filtering
Measurement	nominal value + interval	raw, correction, corrected value + interval
Dynamics	linear model with 4 actions	nonlinear vibration term, flow clamp, U_ctrl, 5 actions
Robustness	nominal simulation	nominal and pessimistic simulation
Decision	best action by outcome and MRE	robust_best_action by worst-case criterion
Verifier	Zig reproduction of nominal procedure	Zig reproduction of nominal and pessimistic procedure

GO-6+ should therefore not be interpreted as a longer version of GO-6. It introduces a new question: can a model remain correct when perceptual uncertainty, decoy signals, corrections and robust analysis are combined into one task?

3. Relation to contemporary AI benchmarks

Contemporary AI benchmarks increasingly emphasize that a single overall score is insufficient to describe a model. HELM is a leading example of holistic evaluation because it advocates coverage across multiple scenarios and metrics rather than a single dominant scale [6].

Multimodal benchmarks such as MMMU show that advanced perception must be connected to domain knowledge and deliberate reasoning rather than only image recognition [7]. GPQA demonstrates that expert-level questions can remain difficult even when evaluators have access to the web, which emphasizes the need for disciplined, verifiable answers [8].

SWE-bench Verified highlights the value of executable verification and human-validated real-world software engineering tasks [9]. GO-6+ adopts this verification principle but moves it into the physical-instrumentation domain: a model must not only write persuasive text, but code that reproduces its procedure.

GO-6+ differs from these benchmarks by combining analog measurement, fuzzy inference, dynamic simulation, robust decision-making, and an executable verifier in one compact but causally connected problem.

4. Formal definition of GO-6+

GO-6+ uses nine images of analog instruments named neutrally from Fig-1.png to Fig-9.png. The file order is not a semantic key. From the visible content, the model must determine the label, unit, scale, reading, interval, and relevance.

Variable	Meaning	Unit	Role in the model
T	process temperature	°C	thermal state; affects pressure and classification
P	process pressure	kPa	mechanical load and input to vibration channel
F	coolant flow	L/min	cooling and pump-starvation condition
V	vibration	mm/s	key instability and catastrophe boundary
D	pressure trend dP/dt	kPa/min	natural direction of pressure change
U_ctrl	regulator control voltage	V	affects R_{eff} and S_{eff}

Three additional instruments are decoy indicators. They may appear technically relevant, but are not part of the defined equations. The model must demonstrate that it does not use signals merely because they look important.

5. Anti-leakage model and evaluation security

GO-6+ uses a strict anti-leakage policy. Leakage is not only an explicit answer. Leakage can be a digital value printed below the needle, a filename that reveals semantic role, metadata containing target values, a JSON schema that pre-populates sensor keys, or an illustration identical to the test input.

The public task statement therefore must not reveal the mapping from Fig-n to model variable. The task may list the set of possible relevant variables, because that is part of the mathematical specification, but it must not reveal which image is the source of which variable.

If a digital readout accidentally appears in an image and directly gives the needle value, the correct model behavior is to report it as a potential leakage artifact and not use it as a replacement for analog pointer reading.

The anti-leakage principle is essential because the benchmark should measure actual perceptual-causal work, not pattern recognition from prior exposure.

6. Analog instrumentation, measurement uncertainty, and calibration

An analog instrument is not a perfect digital sensor. The value is obtained by interpolating the pointer position on a scale. Pointer thickness, viewing angle, image resolution, and scale nonlinearity create a measurement interval. Therefore GO-6+ requires a nominal reading and an interval for every image.

Calibration is the second important component. An instrument may have a fine scale, offset, zero correction, or scale factor. The model must distinguish the raw reading from the corrected value. Downstream computation uses the corrected value.

This is where the difference between describing an image and understanding an instrument becomes visible. A model may read the pointer correctly, but still fail if it ignores calibration or interprets an auxiliary scale as a main sensor.

7. Fuzzy inference

The fuzzy layer is retained because physical state is not binary. A system can be partially normal and partially high in temperature or pressure. Fuzzy sets represent transition zones and provide an initial risk estimate.

```
R1 = min(high(T), high(P))
R2 = min(high(P), rising_fast(D))
R3 = min(low(F), high(T))
R4 = min(unstable(V), high(P))
R5 = max(critical(T), critical(P))

aggregated_risk = max(R1, R2, R3, R4, R5)
```

The fuzzy decision must not be treated as the final action. It detects the current state but does not simulate secondary action consequences. In GO-6+ this is even more important than in GO-6, because `U_ctrl` and the nonlinear vibration term can reverse the intuitive choice.

8. Dynamic model and physical causality

GO-6+ uses a three-tick discrete model. The model is compact but causally dense: temperature, pressure, flow, vibration, control voltage and action parameters propagate into one another.

```
R_eff = R_action + 0.15 * U_ctrl
S_eff = S_action + 0.03 * abs(U_ctrl)

T_next = T + 0.18 * (P - 120) - 0.75 * F + H
P_next = P + D + 0.12 * (T - 65) - R_eff
V_next = V + 0.08 * (P - 125) + S_eff + 0.08 * max(0, V - 6.5)^2
F_next = clamp(F + C, 0.0, 8.5)
```

The nonlinear vibration term is deliberately threshold-based. Below the threshold the system is approximately linear; above it, a small error can substantially change the final outcome. This penalizes rounding, skipped steps and nominal-only reasoning.

9. Nominal, pessimistic, and robust simulation

Nominal simulation uses corrected nominal values. However, a real system does not live at a nominal point; it lives in an uncertainty interval. Therefore GO-6+ requires pessimistic simulation.

The pessimistic simulation uses upper bounds for T, P, V and D, the lower bound for F, and both interval endpoints for U_ctrl. The reason is that U_ctrl has a two-sided effect: larger U_ctrl may support pressure relief, but can worsen vibration stress.

For every action, the worse of the low-U and high-U subcases is selected. Only after that is robust_best_action chosen.

10. Decision protocol and proof of uniqueness

The final decision is not made by following the largest alarm. It is made by a protocol that first eliminates pessimistically catastrophic actions, then prefers actions that are safe in both scenarios, and if none exist chooses the minimum worst-case max_relative_excess.

```
worst_case_max_relative_excess =
max(nominal_max_relative_excess, pessimistic_max_relative_excess)
```

The choice must include a proof of uniqueness. If the difference is less than or equal to 1e-9, intervention_cost is used as tie-break. If the tie remains, unique_best_action must be false.

11. Zig verifier and executable verification

Zig code is part of the methodology, not decoration. The model must produce an executable program that reproduces the numerical procedure. This verifies that the text explanation has operational content.

Required component	Methodological role
State, Action, Scenario, Outcome, SimResult, Choice	clear typing and audit structure
clamp	physical flow saturation
isFiniteState	protection from NaN/Inf and numerical explosions
membership functions	reproduction of fuzzy layer
simulateAction / simulateScenario	reproduction of nominal and pessimistic dynamics
chooseRobustAction	executable proof of decision protocol
RESULTS_JSON + <END>	machine-checkable final trace

12. RESULTS_JSON as an audit trace

RESULTS_JSON must be valid JSON, without placeholders and without empty action objects. It should include instrument_readings, ignored_instruments, model_input_bindings, calibrated_inputs, fuzzy_memberships, rule_activations, nominal_simulation, pessimistic_simulation, action_summary and final decisions.

The JSON is important for semi-automated scoring. Instead of reading the full response subjectively, an evaluator can validate control points and numeric values in a structured output.

13. Scoring matrix and detailed rubric

Area	Points
Instrument identification and decoy filtering	10
Raw analog readings	10
Calibration, corrected values and intervals	10
Model input bindings and units	7
Physical consistency and action conflict	8
Fuzzy inference	10
Nominal dynamic simulation	12
Pessimistic / robust dynamic simulation	13
Decision, robust_best and uniqueness	10
Zig verifier: compilation, accuracy, reliability	8
RESULTS_JSON / formal checkability	2
Total	100

Scoring is additive. The rubric must not be retroactively changed for an individual model. Evaluation should preserve the difference between perceptual errors, numerical errors, logical errors and software errors.

Score range	Interpretation
9.0–10.0	exceptional; nearly all controls pass
8.0–8.9	very strong solution with minor defects
7.0–7.9	good, but with a significant weakness in one major section
6.0–6.9	partially correct; often nominal but not robust
5.0–5.9	superficial; multiple key failures
<5.0	not reliable for GO-6+

Illustrative instruments from an early experimental set

The images in this section are not part of the final Figure_set2 input for GO-6+. They are used only as methodological illustrations: they show the class of analog instruments, earlier problems with ranges, digital leakage or irrelevant values, and must not be used as a reference key for solving the benchmark.



Figure 1. Illustrative analog instruments from an early trial series; not part of the GO-6+ Figure_set2 benchmark input.

The methodological point of this illustration is not to reveal target values, but to show why GO-6+ requires an explicit anti-leakage policy: an instrument can have an unsuitable range, an unnecessary auxiliary scale, a digital readout, or a visual appearance that encourages a model to confuse relevant and decoy channels.

15. Execution protocol and result preservation

For every model, preserve the full raw answer, runtime, token count when available, model version, reasoning mode, whether the images were actually attached, and whether the Zig code compiles.

Separate all-runs tables from best-by-model tables. If a model is run multiple times, preserve every attempt, but ranking may use the best clearly labeled result.

If images, task text or methodology change, the result must be labeled with a new revision. Mixing results from different versions without disclosure is not methodologically valid.

16. Limitations and future extensions

GO-6+ is a methodological test, not a certification-grade industrial standard. The equations are intentionally compact so that they are manually checkable, but connected enough to produce realistic propagation errors.

A future GO-7 can extend this approach to op-amp dynamics, nonlinear oscillators, saturation regimes, hysteresis, delays, chaotic attractors or sensor networks with partial faults.

GO-6+ should remain a bridge: realistic enough to punish pattern-based solving, but controlled enough for the result to be verifiable.

17. Conclusion

GO-6+ Next Ultimate reality introduces a benchmark category in which a model must combine perception, measurement, fuzzy logic, dynamic simulation, robust decision-making and executable code. Its value is not task length, but chain depth. A model that solves GO-6+ demonstrates more than the ability to generate persuasive text: it demonstrates technical discipline, causal control and verifiability.

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Appendix A. Condensed canonical GO-6+ structure

- A-IDENTIFIKACIJA_I_OCITAVANJE_ANALOGNIH_INSTRUMENATA
- B-FIZICKA_KONZISTENTNOST_I_KALIBRACIJA_ULAZA
- C-FUZZY_INFERENCIJA
- D-NOMINALNA_DINAMICKA_SIMULACIJA
- E-PESSIMISTIC_SIMULACIJA_I_ROBUSTNOST
- F-ODLUKA_I_DOKAZ_JEDINSTVENOSTI
- G-ZIG_KOD
- RESULTS_JSON
- <END>

Appendix B. Control points for evaluators

- CP-A: all 9 instruments processed; 6 relevant; 3 decoy; raw/corrected explicit.
- CP-B: physical consistency, U_ctrl role and decoy signals explained.
- CP-C: all fuzzy memberships, R1–R5, aggregated_risk and initial fuzzy decision.
- CP-D: nominal simulation A0–A4, A4 phased rule, R_eff, S_eff, nonlinearity and clamp.
- CP-E: pessimistic scenario, both U_ctrl interval endpoints and worst-case MRE.
- CP-F: robust_best_action, recommended_action, proof of uniqueness.
- CP-G: compilable Zig verifier with real simulation.
- CP-J: valid RESULTS_JSON without placeholders.

14. Detailed evaluation rubric by control component

This rubric allows GO-6+ to be evaluated consistently and separates different classes of errors. A model must not be rewarded merely because it guessed the final action. If it reached the action through a wrong reading, wrong mapping or hardcoded Zig program, the score must reflect the weak evidence chain.

Component	Points	Criterion
Instrument identification and decoy filtering	10	Full score requires processing all nine images, exactly six relevant inputs, exactly three decoy instruments, and a justified semantic map. Using a decoy signal as a model input is a major error.
Raw readings	10	Full score requires reasonable analog reading with an interval and no false precision. A wider interval is better than an unjustified precise value.
Calibration and corrected values	10	Raw reading, correction and corrected value must be separated, and corrected values must be used downstream.
Model input bindings	7	Each of T, P, F, V, D and U_ctrl must have exactly one source, unit, interval and justification.
Physical consistency	8	The model must explain the conflict among pressure, flow, vibration and U_ctrl, and why a fuzzy alarm is not the final action.

Fuzzy inference	10	All membership degrees, R1-R5, aggregated_risk and initial fuzzy decision are required.
Nominal dynamics	12	A0-A4 simulation over three ticks, A4 phased rule, R_eff, S_eff, nonlinearity and clamp are required.
Pessimistic robustness	13	The interval-derived scenario, both U_ctrl endpoints, worse subcase selection and worst-case MRE are required.
Decision and proof	10	nominal_best_action, robust_best_action, recommended_action, elimination proof and uniqueness proof are required.
Zig verifier	8	The code must execute, must not read images, must compute the simulation and output valid JSON.
RESULTS_JSON	2	JSON must be valid, complete and free of placeholders.

15. Taxonomy of typical failures

GO-6+ is useful precisely because failures are not of one type. The following table helps locate the error in the processing chain.

Failure class	Example	Consequence
Perceptual error	Pointer or scale read incorrectly	Wrong input propagates through the chain
Semantic error	Relevant instrument confused with decoy	Wrong input vector is constructed
Calibration error	Offset or fine scale ignored	Nominal solution is systematically shifted
Interval error	Only nominal case computed	No robust decision exists
Fuzzy error	Wrong membership function or boundary	Wrong initial_fuzzy_decision
Dynamic error	Nonlinear term or clamp skipped	Incorrect action classification
Decision error	nominal_best chosen instead of robust_best	Potentially unsafe recommendation
Executable-verification error	Zig does not compile or hardcodes result	No executable proof
Formal error	Invalid JSON or placeholders	Result is not machine-checkable

16. Execution protocol and result preservation

1. Use the same GO-6+.txt and the same Figure_set2 for all models in one comparison.
2. Record the exact model, reasoning mode, runtime, tokens/s if available, total token count and test date.
3. Preserve the full raw model answer without post-editing.
4. Extract the Zig code into a separate file and run it with the target compiler version.

5. Record whether the code compiles on the first attempt, after one correction, or not at all.
6. Score the text chain and the Zig verifier separately, then apply the total rubric.
7. Separate all-runs tables from best-by-model tables.
8. Do not mix GO-6, GO-6+ and later GO-7 results without version labels.

Appendix C. Proposed scoring sheet

Field	Value
Model	
Datum / Date	
GO-6+ text hash	
Figure_set2 hash / archive ID	
Vision input confirmed	
Zig compiler version	
Zig compile status	
Final score / 100	
Score / 10	
Main failure mode	
Reviewer notes	